2022

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Middle Primary Grades 3–4

TIME ALLOWED

60 minutes

Instructions and Information 考试说明

General 一般说明

1. There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 000 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect answer.
   试卷包含 25 道单项选择题和 5 道填空题，填空题答案均为 000-999 之间的整数。题目设置由易到难，答错不扣分。
2. You may use any teaching aids normally available in your classroom, such as MAB blocks, counters, currency, calculators, play money etc. You are allowed to work on scrap paper and teachers may explain the meaning of words in the paper. Mobile phones, computers or tablets are not permitted.
   考生允许携带简单无编程计算器和常用教具，例如：算数积木、筹码、货币、玩具钞票、便条纸等，监考老师可以为考生解释试卷上出现的单词含义，考生不允许携带手机、电脑或者平板。
3. Diagrams are NOT drawn to scale. They are intended only as aids.
   图形只作为辅助参考，并未按照实际比例绘制。
4. This is not a test so do not worry if you can't answer all the questions. However, try to answer as many as you can — you do not lose marks for incorrect answers.
   这不是一场普通的测验，所以即使不能回答所有问题也不必担心。不过要尽可能回答更多的题目，答错也不会丢掉分数的！
5. Check the details on your answer sheet carefully.
   务必仔细检查答题卡上填写的所有信息。
6. When your teacher gives the signal, begin working on the problems.
   监考老师发出作答指令后，方可开始答题。

Integrity of the competition

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

Reminder

You may sit this competition once, in one division only, or risk no score.

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The answer sheet 答题卡说明

Your answer sheet will be scanned. To make sure the scanner reads your paper correctly, there are some DOs and DON'Ts:

你的答题卡将通过扫卡机进行扫描，为确保扫卡机准确扫描你的答案，请务必遵守以下“六要”和“二不要”：

DO 要：

- Mark your exam code and answers with a **2B** pencil.
  使用 **2B** 铅笔正确填涂准考证号和答案。
- Record your answers on the answer sheet (not on the question paper).
  将答案全部涂在答题卡上（在试卷上作答无效）。
- For questions 1–25, fully colour the bubble matching your answer.
  将题目 1–25 的答案按照填涂示例正确涂在对应方格中。
- For questions 26–30, write your answer in the boxes and fully colour the bubble mathching your answer.
  将题目 26–30 的答案写在方框中，并按照填涂示例正确涂在对应方格中。
- Use an eraser if you want to change an answer or remove any marks or smudges.
  请保持答题卡清洁，修改时用橡皮擦干净。
- Check the number of the answer you are filling in is the same as the number of the question you are answering.
  检查答题卡上填涂的答案数量和作答的题目数量是否一致。

DO NOT 不要：

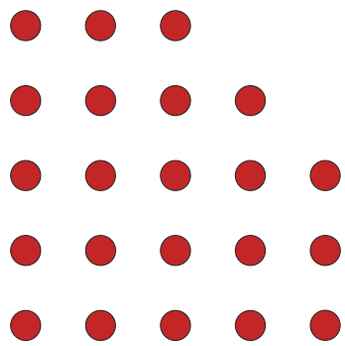
- Doodle or write anything extra on the answer sheet.
  不要在答题卡上乱写乱画。
- Go outside the red boundaries of the answer sheet.
  不要超出答题卡红色虚线框。

Middle Primary Division

1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

1. 请问右图中一共有多少个点？

How many dots are in this pattern?



(A) 20 (B) 21 (C) 22 (D) 23 (E) 24

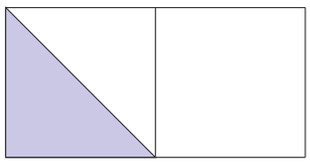
2. 请问比 465 大一百的数是多少？

What number is one hundred more than 465?

(A) 365 (B) 455 (C) 475 (D) 565 (E) 1465

3. 右图中，请问阴影部分占矩形面积的几分之几？

What fraction of this rectangle is shaded?

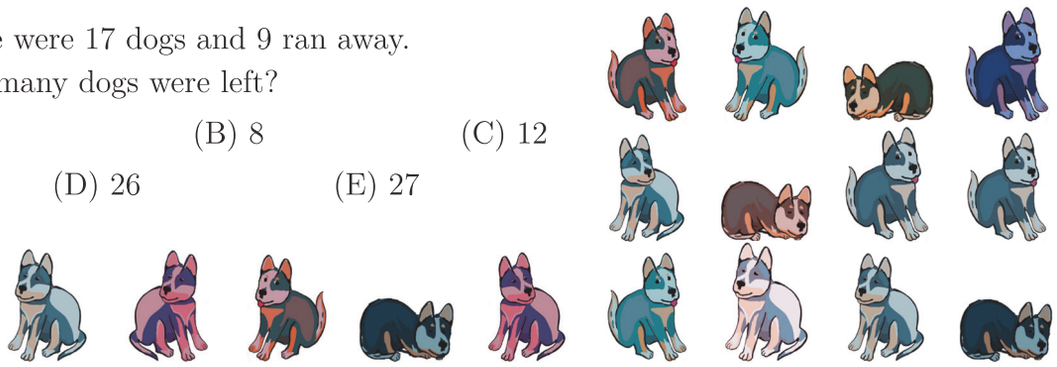


(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{8}$ (E) $\frac{1}{10}$

4. 这里一共有 17 只狗，其中 9 只跑走了。
请问还剩下多少只狗？

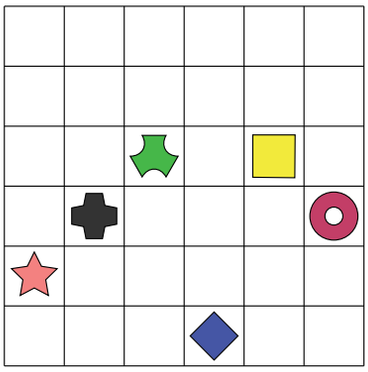
There were 17 dogs and 9 ran away.
How many dogs were left?

(A) 7 (B) 8 (C) 12 (D) 26 (E) 27



5. 约翰在玩棋盘游戏。
他将蓝色方块 ◆ 向上移动一个小方格再向左移动三个小方格。
请问他会走到哪个图形上？

John is playing a board game.
He moves his blue piece ◆ one square up then three squares left.
Which piece does he land on?



(A) ★ (B) ✚ (C) ✱ (D) ■ (E) ◎

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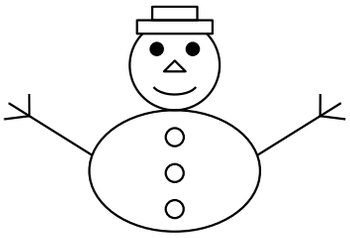
6. 请问右图中的雪人**没有**用到哪一种形状？
(A) 圆形 (B) 椭圆 (C) 三角形 (D) 正方形 (E) 长方形

Which shape is **not** used in this snowman picture?
(A) circle (B) oval (C) triangle (D) square (E) rectangle



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7. 小伊从 20 开始数数，每次增加 2：20、22、24、$\cdots$。请问她数到的第十个数是多少？

Eve starts at 20 and counts up by twos: 20, 22, 24 and so on. What is the tenth number she counts?
(A) 30 (B) 32 (C) 34 (D) 36 (E) 38

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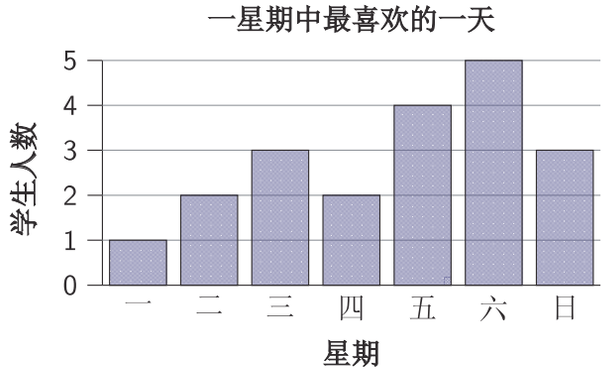
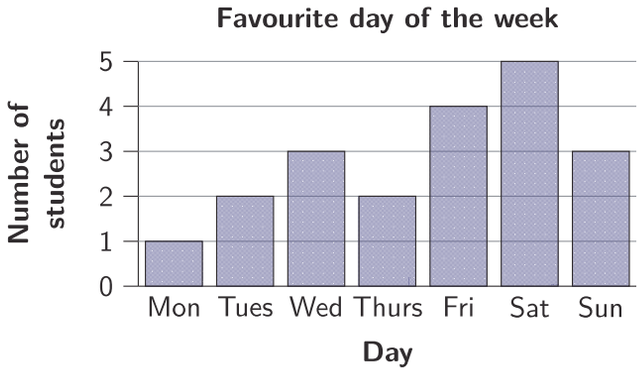
8. 右图是 3 年级学生制作的。
请问一共有多少名学生最喜欢的一天是星期六或者星期日？

(A) 5 (B) 8 (C) 10 (D) 12 (E) 20



This graph was made by a Year 3 class.
How many students chose either Saturday or Sunday as their favourite day?

(A) 5 (B) 8 (C) 10 (D) 12 (E) 20



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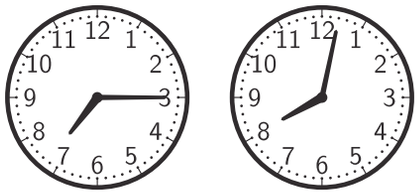
9. 今天早上我骑自行车兜风。
右侧时钟显示了我骑行开始和结束的时间。
请问我一共骑行了多少分钟？

I went for a bike ride this morning.
These clocks show my start and finish times.
In minutes, how long was my ride?



开始时间 Start　　结束时间 Finish

(A) 9 (B) 13 (C) 47 (D) 52 (E) 62

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10. 小伊、小露与其他小孩站成一列。从前往后数，小伊站在第四位；从后往前数，小露站在第四位。如果这列队伍一共有 15 个小孩，请问小伊与小露之间有多少个小孩？

Edie and Louie are standing in a line with other children. Edie is fourth from the front and Louie is fourth from the back of the line. There are 15 children in the line. How many children are between Edie and Louie?

(A) 7 (B) 8 (C) 10 (D) 11 (E) 12

11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

11. 我的粉色小猪存钱罐内有 49 枚 10¢ 的硬币，蓝色小猪存钱罐内有 25 枚 20¢ 的硬币。
请问我一共有多少钱？（1\$ = 100¢）

There are 49 ten-cent coins in my pink piggy bank and 25 twenty-cent coins in my blue piggy bank.
How much money do I have altogether?

(A) \$7.40 (B) \$9.90 (C) \$12.30 (D) \$14.80 (E) \$990

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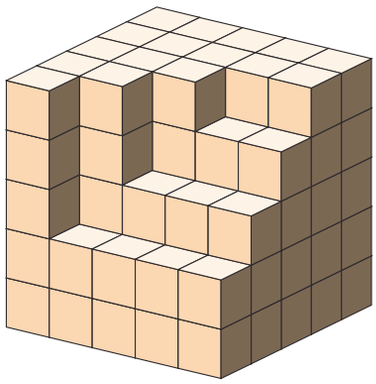
12. 请问还需要多少个单位立方体才能完成这个大立方体？

How many more small cubes are needed to complete this large cube?



(A) 6 (B) 8 (C) 9 (D) 12 (E) 20

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13. 阿美老师带了 100 张贴纸分给班上的 23 位学生，每位学生都分得数量相同的贴纸。请问她最少会剩下多少张贴纸？

Ms Amali brings 100 stickers to share equally among her class of 23 students.
How many stickers will she have left over?

(A) 3 (B) 8 (C) 12 (D) 17 (E) 21

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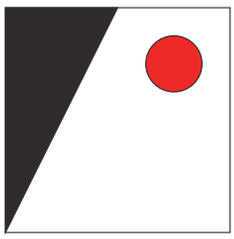
14. 将这张卡片沿着它的右侧边缘翻转，然后再沿着底部边缘翻转。
请问这张卡片现在是什么样子的？

This card is flipped over its right-hand edge and then flipped again over its bottom edge.
What does the card look like now?



(A)
(B)
(C)
(D)
(E)

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15. 小可打算在两个方格内填入相同的数字使下面等式成立。请问她应该填入哪个数字？

$$7 + \square = 11 - \square$$

Chris wants to use the same number in both boxes to make this number sentence true. What number should she use?

$$7 + \square = 11 - \square$$

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

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16. 小莎对哥哥说：「你比我大 10 岁。」她的哥哥回答：「你说的没错，而且我的年龄是你的三倍。」请问小莎多少岁？

Sally says to her brother, 'You are 10 years older than me'. Her brother says, 'You are right, and I am three times your age'. How old is Sally?

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

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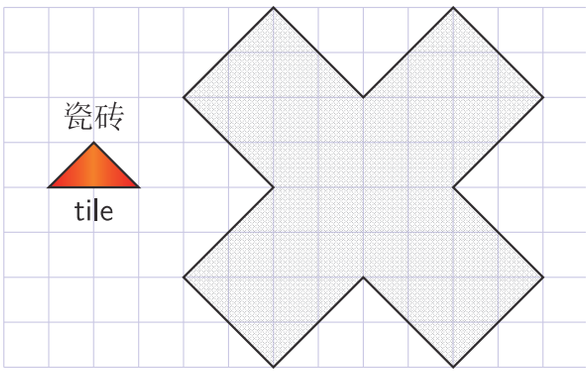
17. 请问一共需要用多少块彩色小瓷砖才能将右图所示的十字图形铺满？

How many of the small coloured tiles are needed to cover the large cross shape shown?

(A) 10 (B) 15 (C) 20 (D) 36 (E) 40



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18. 小舒有编号 1 到 7 的卡片。
她将其中六张卡片放入右侧的三个等式中并使等式成立。
请问会留下哪一张卡片？

Siobhan has seven cards, numbered 1 to 7.
She places six of them correctly in the three number sentences.
Which card does she have left over?

(A) 1 (B) 2 (C) 3 (D) 5 (E) 7

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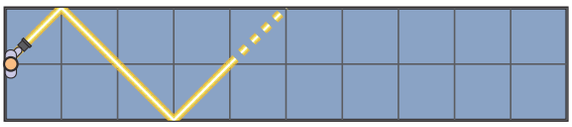
19. 一间长为 10 m、宽为 2 m 的矩形房间中四面墙上都有镜子。任何射向这些镜子的光线都会以相同的角度反射出去，如下图所示。
一名警卫站在房间的一端打开手电筒以 45° 的方向射向墙壁，发出的一道光束在镜子之间反射多次，直到光束反射回警卫处才停止。
请问这束光在镜子之间一共反射了多少次？



A long rectangular room 10 m long and 2 m wide has mirrors on all four walls. Any beam of light hitting these mirrors bounces back at the same angle as shown below.
A guard standing at one end of the room shines a torch at an angle of 45° to the walls, making a narrow beam that bounces off the mirrors several times, stopping when it returns to her.
How many times does the beam of light bounce off the mirrors?

(A) 7 (B) 9 (C) 10 (D) 11 (E) 22

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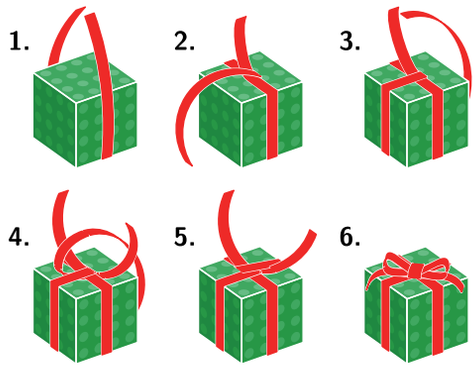
20. 一个装有 30 枚珠子的罐子重 115 g，装有 20 枚珠子的相同罐子重 85 g。请问空罐子的重量是多少？

A can filled with 30 marbles weighs 115 g. The same can with 20 marbles weighs 85 g. How much does the empty can weigh?

(A) 10 g (B) 20 g (C) 25 g (D) 30 g (E) 55 g

21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 彼得打算购买一条彩带用右图所示的方式来包装礼盒。
这个礼盒长 20 cm，宽 20 cm，高 20 cm。
另外还需要 80 cm 长的彩带制作蝴蝶结。
为了包装这个盒子，请问彼得需要购买彩带的最佳长度大约是多少？



Peter wants to buy a length of ribbon to wrap around a box as shown.
The box is 20 cm long, 20 cm wide and 20 cm high. It takes an extra 80 cm of ribbon to tie the bow.
What is the best estimate of the amount of ribbon that Peter needs to buy to tie around the box?

(A) 160 cm (B) 180 cm (C) 240 cm (D) 280 cm (E) 320 cm

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22. 在第 1 周时，小汉与小伊在银行开帐户存款。从第 1 周开始，小汉每两周存入 \$12，小伊在第 1 周存入 \$32 且之后每一周都存入 \$4。请问两人在银行的存款首次相同时是第几周？

(A) 第 3 周　(B) 第 5 周　(C) 第 7 周　(D) 第 9 周　(E) 第 11 周

In week 1, Hamish and Eliza open bank accounts for their savings. Hamish saves \$12 every two weeks, starting in week 1. Eliza saves \$32 in week 1 and then \$4 every week after that. When will they first have the same amount of money in the bank?

(A) week 3　(B) week 5　(C) week 7　(D) week 9　(E) week 11

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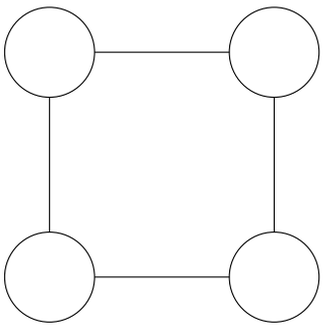
23. 在正方形四个角的圆圈中填入互不相同的正整数。同一条边相连的两个数之差都大于 1。
请问这四个数相加之和的最小可能值是多少？

A different positive whole number is placed in each corner of a square. Two numbers joined by an edge must have a difference of more than 1.
When the four numbers are added together, what is the smallest possible total?

(A) 10　(B) 11　(C) 12　(D) 13　(E) 14



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24. 三个正整数相加之和为 21，相乘之积为 280。请问这三个数中最小的数是多少？

Three positive whole numbers add to 21. When these same three numbers are multiplied together they equal 280. What is the smallest of these three numbers?

(A) 1　(B) 2　(C) 4　(D) 7　(E) 10

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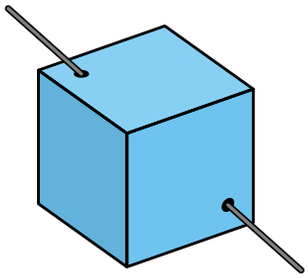
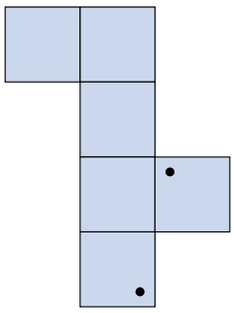
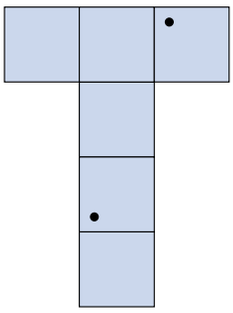
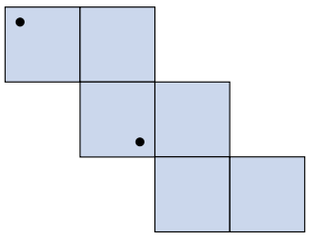
25. 小狄用立方体的展开图折叠成右图所示的立方体，然后用一根针穿过这个立方体。
抽出这根针后，立方体上留下了两个孔。再将这个立方体展开，形成原来的展开图。请问下列哪个选项可能是这个立方体的展开图？

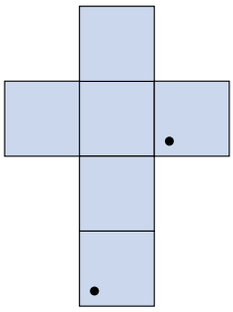


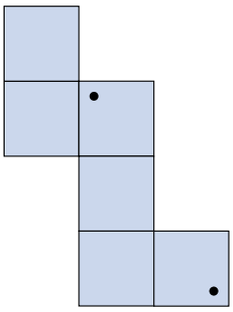
Steven made this cube from a paper net, then pushed a pin through it as shown. He then removed the pin, leaving holes in the cube, and unfolded the cube back to its net.
Which of the following could be the net of Steven's cube?

(A)


(B)


(C)


(D)


(E)


问题 26-30 的答案为 000-999 之间的整数，
请将答案填在答案卡上对应的位置。
第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，
第 29 题占 9 分，第 30 题占 10 分。

For questions 26 to 30, the answer is an integer from 000 to 999.
Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

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26. 我观察到电表目前的读数是 896754 度，各个数位上的数字都不相同。请问我最少要再使用多少度电才能使得各个数位上的数字再次互不相同？

I notice that my electricity meter currently reads 896 754 units, where all the digits are different. How many more units of electricity will I need to use before all the digits are again different?

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27. 请问在 200 与 500 之间总共有多少个整数含有数字 3?

How many whole numbers between 200 and 500 contain the digit 3?

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28. 如图所示，一个数字显示器使用可以发光的小条来组合成数字：

对于某些特别的数，组成它的发光小条的数量恰好与它各个数位上的数字之和相等。例如，373 这个数字是由 $5+3+5=13$个发光的小条组成的，刚好等于它的各个数位上的数字之和 $3+7+3=13$。
请问满足这样条件的最大三位数是多少？
On a digital display, a combination of bars light up to represent each digit as shown. In some special numbers, the number of bars which light up in the digits is the same as the sum of the digits. For example, in 373 the number of bars is $5+3+5=13$ which is the equal to $3+7+3=13$. What is the largest such three-digit number?

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29. 小阮根据以下规则写下了一些数。
从 1 开始，他将这个数翻倍再加 4，因此他写下的第二个数为 6。
接着他重复这个过程，从最后写下的数开始将它翻倍再加 4，但是如果得到的数大于 100，则不写出这个数的百位数。
请问小阮写下的第 2022 个数是多少？

Nguyen writes down some numbers according to the following rules.
Starting with the number 1, he doubles the number and adds 4, so the second number he writes is 6.
He now repeats this process, starting with the last number written, doubling and then adding 4, but he doesn't write the hundreds digit if the number is bigger than 100.
What is the 2022nd number that Nguyen writes down?

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30. 我从 1、3、5、7、9、 . . . 、 105 中选出 3 个互不相同的数，然后将它们相加。

请问我一共可以得到多少个不同的和？

I choose three different numbers out of this list and add them together:

$$1, 3, 5, 7, 9, \ldots, 105$$

How many different totals can I get?